Coding and Gambling:
Two Vocabularies for the same theory

(After Chapters 5 and 6 of Cover and Thomas)

Lecture notes for an information theory course
Instituto de Matemática e Estatística, Universidade de São Paulo

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Notation and standing conventions

Logarithms are base 2 and information is measured in bits; $\log_D$ is written when the base is the code alphabet size D. Prerequisites from Chapters 2 to 4 of Cover and Thomas [1] are used freely: entropy $H(X)$, conditional entropy $H(X|Y)$, relative entropy $D(p\,\|\,q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$ with $D(p\,\|\,q) \geq 0$ and equality iff $p = q$, mutual information $I(X;Y) = H(X) - H(X|Y)$, the chain rule, the entropy rate of a stationary process, and the weak law of large numbers. The sessions on Chapters 5 and 6 of this course (the "compression lectures") are required background, but everything used here is stated and proved again.

Symbol	Meaning (where defined)
$\mathcal{X},\ m,\ x$	source alphabet, its size $m = \lvert\mathcal{X}\rvert$, a source symbol; in the race, the m horses and the winner (Lecture 1)
$p(x),\ p_i$	probability that $X = x$, or that horse i wins (Lecture 1)
$\mathcal{D},\ D,\ \mathcal{D}^*$	code alphabet $\{0, \dots, D-1\}$, its size, the finite strings over it (Lecture 1)
$C(x),\ l(x),\ L$	codeword of x, its length, and the expected length $\sum_x p(x)l(x)$ (Lecture 1)
c	a Kraft sum: $\sum_x D^{-l(x)}$ for a code, $\sum_i 1/o_i$ for a set of odds (Lectures 1 and 2)
$l_q(x) = \log \frac{1}{q(x)}$	ideal length of x under the model q; real-valued, not necessarily an integer (Lecture 1)
$o_i,\ b_i$	odds, quoted o_i-for-1, and the fraction of wealth bet on horse i (Lecture 1)
$S(x) = b(x)o(x)$	wealth relative: factor by which wealth is multiplied when x wins (Lecture 1)
$W(b,p),\ W^*(p)$	doubling rate $\mathbb{E}\log S(X)$ of the bet b under p, and its maximum over b (Lecture 1)
$l_o(x) = \log o(x)$	the bookmaker's code: length that the odds assign to x (Lecture 2)
$l_b(x) = \log \frac{1}{b(x)}$	the gambler's code: ideal length that the bet assigns to x (Lecture 2)
$r_i = \frac{1/o_i}{c}$	the bookmaker's implied probability of horse i (Lecture 2)
$x^n,\ b(x^n),\ S_n$	a sequence $(x_1, \dots, x_n)$, the wealth fraction bet on it, the wealth after n races (Lecture 3)
$b(x_k \mid x^{k-1})$	sequential bet on the kth outcome given the past (Lecture 3)
$F(x),\ \overline{F}(x)$	cumulative distribution of the bet in lexicographic order and its midpoint version (Lecture 3)
$\lfloor F \rfloor_l$	the number $F \in [0,1)$ truncated to its first l binary digits (Lecture 3)
$b(x \mid y),\ l(x \mid y)$	conditional bet and conditional codeword length given side information y (Lecture 4)
$i(x;y) = \log \frac{p(x\mid y)}{p(x)}$	pointwise mutual information; $\mathbb{E}\, i(X;Y) = I(X;Y)$ (Lecture 4)
$h(q)$ or H(p)	binary entropy function $-q\log q - (1-q)\log(1-q)$ (Lecture 4)

Convention 0.1 *(Standing conventions)***.**

(i) The source alphabet $\mathcal{X}$ is finite and every symbol has strictly positive probability; a symbol of probability 0 is dropped. The same symbol x names a source symbol and a horse, and $m = \lvert\mathcal{X}\rvert$ is both the alphabet size and the number of horses.

(ii) A *bet* b is a probability mass function on $\mathcal{X}$: $b(x) \geq 0$ and $\sum_x b(x) = 1$. The gambler reinvests all wealth in every race; keeping cash aside is discussed where it matters.

(iii) Odds are quoted a-for-1: one unit staked returns a units, stake included, if the horse wins and nothing otherwise. Odds are strictly positive and finite.

(iv) "Optimal code" means a prefix code of minimum expected length for the given distribution; by McMillan's theorem, allowing all uniquely decodable codes changes nothing.

(v) A bet $b(x) = 0$ on an outcome of positive probability gives doubling rate $-\infty$: ruin.

(vi) $x^n = (x_1, \dots, x_n)$ is a sequence and x^{k-1} its prefix of length $k-1$; x^0 is the empty sequence. Sequences of the same length are ordered lexicographically.

1 Two problems, one inequality

A source coder is handed a random variable X and must choose, for each value x, a string of code symbols. A horse race gambler must choose, for each horse x, a fraction of wealth to bet on it. Assume that the probabilities of the outcomes are known to the coder and the gambler and that in both problems the natural scores, expected codeword length and expected logarithmic growth, are optimized by the same choice and bounded by the same quantity. This lecture sets up the two problems in parallel and proves their two basic theorems from one identity.

1.1 The coding problem

Definition 1.1 *(Source code, prefix code, expected length).* Let X take values in a finite set $\mathcal{X}$, the *source alphabet*, with probability mass function p, and let $\mathcal{D} = \{0, 1, \dots, D-1\}$ be a *code alphabet* with $D \geq 2$ symbols. A *source code* is a map $C : \mathcal{X} \to \mathcal{D}^*$ into the finite strings over $\mathcal{D}$; the string $C(x)$ is the *codeword* of x and $l(x)$ is its length. The code is a *prefix code* if no codeword is a prefix of another codeword. Its *expected length* is

$$L(C) = \sum_{x \in \mathcal{X}} p(x)\, l(x). \tag{1}$$

A prefix code can be decoded symbol by symbol: as soon as the received string matches a codeword, that codeword is complete, because no longer codeword starts the same way.

The lengths of a prefix code cannot all be short. The constraint is the following inequality.

Theorem 1.2 *(Kraft inequality).* *The lengths $l(x)$, $x \in \mathcal{X}$, of a prefix code over a D-ary alphabet satisfy*

$$c := \sum_{x \in \mathcal{X}} D^{-l(x)} \leq 1. \tag{2}$$

Conversely, for any positive integers $l(x)$ satisfying (2) there is a prefix code with these lengths.

Proof. Associate with a string $w = y_1 y_2 \cdots y_l$ over $\mathcal{D}$, that is, represent w by the number $0.y_1 y_2 \cdots y_l$ in base D.

The half-open interval given by

$$I(w) = \Big[\, 0.y_1 y_2 \cdots y_l,\ 0.y_1 y_2 \cdots y_l + D^{-l} \Big) \subseteq [0, 1), \qquad 0.y_1 \cdots y_l := \sum_{j=1}^{l} y_j D^{-j},$$

is the set of numbers whose D-ary expansion begins with the digits of w. Its length is D^{-l}. The intervals of all D^l strings of length l partition $[0, 1)$. If w is a prefix of w' then $I(w') \subseteq I(w)$. If neither of w, w' is a prefix of the other, say with $l(w) \leq l(w')$, then the first $l(w)$ digits of w' form a string $w'' \neq w$ of the same length as w, so $I(w'')$ and $I(w)$ are distinct cells of one partition and are disjoint, and $I(w') \subseteq I(w'')$ is disjoint from $I(w)$. For a prefix code the intervals $I(C(x))$ are therefore pairwise disjoint subsets of $[0, 1)$, and the sum of their lengths, which is c, is at most 1.

Conversely, list the required lengths in increasing order $l_1 \leq l_2 \leq \cdots \leq l_m$ and put $F_0 = 0$, $F_i = \sum_{j \leq i} D^{-l_j}$. Every D^{-l_j} with $j < i$ is an integer multiple of D^{-l_i}, so F_{i-1} is a multiple of D^{-l_i} and has an exact D-ary expansion w_i with l_i digits. Then $I(w_i) = [F_{i-1}, F_i)$. These intervals are disjoint because $F_m = c \leq 1$, hence no w_i is a prefix of another w_j, and $\{w_i\}$ is the required prefix code. □

Remark 1.3 *(Uniquely decodable codes).* A code is *uniquely decodable* if every finite concatenation of codewords can be parsed in only one way. McMillan proved that the lengths of any uniquely decodable code also satisfy (2) [1, Theorem 5.5.1]. Since the converse half of Theorem 1.2 then produces a prefix code with the same lengths, nothing is gained by leaving the class of prefix codes, and "optimal code" means optimal prefix code throughout.

The quantity $D^{-l(x)}$ behaves like a probability: it is positive and, by Kraft, it sums to at most one. This suggests reading any positive function on $\mathcal{X}$ as a set of lengths.

Definition 1.4 *(Ideal length under a q-model).* A *q-model* is a function $q : \mathcal{X} \to (0, \infty)$. The *ideal length* of x under q is

$$l_q(x) = \log \frac{1}{q(x)} \qquad (\text{bits; } \log_D \frac{1}{q(x)} \text{ in } D\text{-ary symbols}). \tag{3}$$

Conversely a code with lengths $l(x)$ has the *implied q-model* $q(x) = D^{-l(x)}$, whose total mass is the Kraft sum c.

Ideal lengths are real numbers. They become codeword lengths after rounding up, at a cost of less than one symbol each, and Corollary 1.10 below records what that costs on average. Working with real lengths first and rounding last is convenient to make the correspondence with gambling clear.

1.2 The gambling problem

Definition 1.5 *(Horse race, odds, bet, wealth relative).* A *horse race* has m horses labelled by $\mathcal{X}$; horse x wins with probability $p(x)$ and pays *odds* $o(x)$-for-1: one unit staked on x returns $o(x)$ units, stake included, if x wins, and nothing if it loses. A *bet* is a probability mass function b on $\mathcal{X}$: the gambler stakes the fraction $b(x)$ of current wealth on horse x, for every x. If horse X wins, wealth is multiplied by the *wealth relative*

$$S(X) = b(X)\, o(X). \tag{4}$$

If the race is run again and again with independent outcomes $X_1, X_2, \dots$ and the gambler reinvests everything, the wealth after n races, starting from one unit, is the product $S_n = \prod_{k=1}^{n} S(X_k)$. A product of many random factors is controlled by the average of their logarithms, so the score that matters is the following.

Definition 1.6 *(Doubling rate).* The *doubling rate* of the bet b against the win distribution p at odds o is

$$W(b,p) = \mathbb{E} \log S(X) = \sum_{x \in \mathcal{X}} p(x) \log\bigl(b(x)o(x)\bigr), \tag{5}$$

and the *optimal doubling rate* is $W^*(p) = \max_b W(b,p)$, the maximum over all bets.

Theorem 1.7 *(Wealth grows at the doubling rate).* *Let $X_1, X_2, \dots$ be independent with distribution p and let $b(x) > 0$ for all x. Then*

$$\frac{1}{n} \log S_n \;\longrightarrow\; W(b,p) \qquad \text{in probability.}$$

In words, $S_n = 2^{\,nW(b,p)}$ to first order in the exponent: the gambler doubles the wealth W times per race on average.

Proof. $\frac{1}{n} \log S_n = \frac{1}{n} \sum_{k=1}^{n} \log S(X_k)$ is the average of n independent copies of the bounded random variable $\log S(X)$, whose mean is $W(b,p)$. The weak law of large numbers gives the claim. If $b(x) = 0$ for some x with $p(x) > 0$, that horse eventually wins and the gambler is ruined, in agreement with $W = -\infty$. □

The theorem says which quantity the gambler should maximize. It is not the expected wealth after one race: staking everything on the horse with the largest $p(x)o(x)$ maximizes $\mathbb{E}S(X)$ but leads to ruin the first time that horse loses, which happens with probability one in the long run.

1.3 One identity, two theorems

Consider the following elementary computation.

Lemma 1.8 *(Length identity).* *Let p be a probability mass function on $\mathcal{X}$ and $q : \mathcal{X} \to (0, \infty)$ a q-model with total mass $c = \sum_x q(x)$, and let $\tilde{q} = q/c$ be the normalized q-model. The expected*

ideal length of $X \sim p$ under q is

$$\mathbb{E}_p\, l_q(X) = \sum_x p(x) \log \frac{1}{q(x)} = H(p) + D(p \,\|\, \tilde{q}) + \log \frac{1}{c}. \tag{6}$$

Proof. For each x, $\log \frac{1}{q(x)} = \log \frac{1}{p(x)} + \log \frac{p(x)}{\tilde{q}(x)} - \log c$, because $q = c\tilde{q}$. Multiply by $p(x)$ and sum. The three sums are $H(p)$, $D(p \,\|\, \tilde{q})$ and $-\log c$. □

Read the identity as a budget, that is, the amount of money you have to spend with each observation of X.

Describing X with the q-model q costs the unavoidable $H(p)$, plus a penalty $D(p \,\|\, \tilde{q}) \geq 0$ for pointing the q-model in the wrong direction, plus a term $\log(1/c)$ for its total mass: a q-model of mass less than one wastes $\log(1/c)$ bits, a q-model of mass greater than one is too good to be a code and the term is negative.

Theorem 1.9 *(Entropy bound for codes).* *For any prefix code over a D-ary alphabet, $L(C) \geq H_D(X)$, where $H_D(X) = H(X)/\log D$ is the entropy in D-ary units. Equality holds if and only if $l(x) = \log_D \frac{1}{p(x)}$ for every x, that is, if and only if $p(x) = D^{-l(x)}$.*

Proof. Apply Lemma 1.8 with logarithms in base D to the implied q-model $q(x) = D^{-l(x)}$, whose ideal lengths are the actual lengths $l_q(x) = l(x)$ and whose mass is the Kraft sum c:

$$L(C) = H_D(p) + D_D(p \,\|\, \tilde{q}) + \log_D \frac{1}{c} \;\geq\; H_D(p),$$

since $D_D(p \,\|\, \tilde{q}) \geq 0$ and $c \leq 1$ by Theorem 1.2. Equality forces $c = 1$ and $p = \tilde{q} = q$. □

Corollary 1.10 *(Shannon code).* *Let q be a probability mass function on $\mathcal{X}$ and let $l(x) = \lceil \log \frac{1}{q(x)} \rceil$. These lengths satisfy Kraft, so they are the lengths of a prefix code, and when $X \sim p$,*

$$H(p) + D(p \,\|\, q) \;\leq\; \mathbb{E}_p\, l(X) \;<\; H(p) + D(p \,\|\, q) + 1. \tag{7}$$

In particular the code built from the true distribution, $q = p$, has $H(X) \leq L < H(X) + 1$, and an optimal code can only be shorter.

Proof. $\sum_x 2^{-\lceil \log(1/q(x)) \rceil} \leq \sum_x q(x) = 1$, so Kraft holds. Since $\log \frac{1}{q(x)} \leq l(x) < \log \frac{1}{q(x)} + 1$, taking expectations under p and applying Lemma 1.8 to q, which has mass 1, gives (7). □

The corollary is Theorem 5.4.3 of the book: believing that the distribution is q when it is p costs $D(p \,\|\, q)$ bits per symbol, up to the rounding bit. It is the coding half of a statement whose gambling half is the next theorem.

Theorem 1.11 *(Proportional betting is log-optimal).* *For any odds o and any bet b,*

$$W(b, p) = \sum_x p(x) \log o(x) - H(p) - D(p \,\|\, b). \tag{8}$$

Hence $W^*(p) = \sum_x p(x) \log o(x) - H(p)$, attained by the proportional *(or Kelly) bet $b^* = p$ and by no other bet.*

Proof. $W(b, p) = \sum_x p(x) \log o(x) - \sum_x p(x) \log \frac{1}{b(x)}$, and the second sum is $\mathbb{E}_p l_b(X) = H(p) + D(p \,\|\, b)$ by Lemma 1.8, since b has mass 1. Maximizing means minimizing $D(p \,\|\, b)$, which is 0 exactly at $b = p$. □

The optimal bet does not depend on the odds: The odds enter only through the constant $\sum_x p(x) \log o(x)$, which is the expected ideal length of X under the q-model $1/o$, and the identity applies to that q-model too.

Corollary 1.12 *(Doubling rate as a difference of divergences).* *Let $c = \sum_x 1/o(x)$ and $r(x) = \frac{1/o(x)}{c}$, a probability mass function. Then for every bet b,*

$$W(b,p) = D(p \,\|\, r) - D(p \,\|\, b) - \log c, \qquad W^*(p) = D(p \,\|\, r) - \log c. \tag{9}$$

Proof. Apply Lemma 1.8 to $q = 1/o$: $\sum_x p(x) \log o(x) = H(p) + D(p \,\|\, r) + \log \frac{1}{c}$. Insert in (8). □

Theorem 1.13 *(Conservation law).* *At uniform odds $o(x) = m$ for all x,*

$$W^*(p) + H(p) = \log m. \tag{10}$$

Proof. Here $c = 1$ and r is uniform, so $D(p \,\|\, r) = \log m - H(p)$ in (9). □

The conservation law is the first quantitative statement of the relation between the two problems.

On the **coding** side, $\log m$ bits describe X without using its distribution, $H(p)$ is the ideal expected length when the distribution is used, and the difference is **what compression saves**.

On the **gambling** side that same difference is the **doubling rate**.

1.4 Worked examples

Example 1.14 *(The running example: a dyadic source both as a code and as a race).* Let $\mathcal{X} = \{a, b, c, d\}$ with $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$, so $H(p) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 1.75$ bits.
As a code. The binary prefix code $C(a) = \texttt{0}$, $C(b) = \texttt{10}$, $C(c) = \texttt{110}$, $C(d) = \texttt{111}$ has lengths $(1, 2, 3, 3)$, Kraft sum $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 1$ and expected length $L = 1.75 = H(p)$. It is optimal by Theorem 1.9, because $p(x) = 2^{-l(x)}$ for every x. A code that ignores the distribution uses $\log m = 2$ bits for every symbol.
As a race. Four horses at uniform odds 4-for-1. The Kelly bet is $b = p$ and the wealth relatives are $S(x) = 4p(x) = (2, 1, \frac{1}{2}, \frac{1}{2})$, so $\log S(x) = (1, 0, -1, -1)$. The doubling rate is $W^* = \frac{1}{2}(1) + \frac{1}{4}(0) + \frac{1}{8}(-1) + \frac{1}{8}(-1) = 0.25$ bits per race, which is $\log 4 - H(p) = 2 - 1.75$ as Theorem 1.13 says. Over 100 races the wealth is $2^{25} \approx 3.4 \times 10^7$ to first order in the exponent.
The two columns agree symbol by symbol. The numbers $\log S(x) = (1, 0, -1, -1)$ are exactly $2 - l(x)$, the number of bits the prefix code saves on x against the 2-bit description. Horse a doubles the wealth and symbol a saves one bit; horses c, d halve the wealth and symbols c, d cost one bit more than the fixed-length code. What the example shows: at uniform odds, wealth relative and bits saved are the same table, entry by entry, and not only on average.

Example 1.15 *(Three horses at 3-for-1).* Three horses with $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, odds 3-for-1 on each. These are fair odds for a bookmaker who believes the horses equally likely. $H(p) = 1.5$ bits and $\log 3 = 1.5850$, so by Theorem 1.13 the Kelly bet $b = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ has doubling rate $W^* = \log 3 - 1.5 = 0.0850$ bits per race. The wealth relatives are $S = (1.5, 0.75, 0.75)$: the favourite wins half the time and multiplies wealth by 1.5, the others lose a quarter. After 100 races the wealth is about $2^{8.50} \approx 362$ to first order in the exponent. The tempting alternative, everything on the favourite, has $W = \frac{1}{2} \log 3 + \frac{1}{2} \log 0 = -\infty$ and is ruined at the first loss.
The same numbers on the coding side. The Shannon code for p has lengths $\lceil \log(1/p(x)) \rceil = (1, 2, 2)$, for instance $\texttt{0}, \texttt{10}, \texttt{11}$, with expected length $1.5 = H(p)$. Describing 100 independent outcomes with it costs 150 bits on average. Describing them without using p costs $100 \log 3 = 158.50$ bits, rounded up to 159, the length of an index into the 3^{100} possible sequences. The saving before rounding, 8.5 bits, is $100\, W^*$. What the example shows: the reference against

which compression is measured is $\log m$ bits per symbol even when $\log m$ is not an integer, because a fixed-length description of long blocks costs $\log m$ per symbol up to one bit in total.

1.5 The relation

Coding	Gambling
alphabet $\mathcal{X}$, m symbols	m horses
distribution p of the source	win probabilities p
lengths $l(x)$, Kraft sum $c \le 1$	bet $b(x)$, $\sum_x b(x) = 1$
optimal lengths $\log \frac{1}{p(x)}$	optimal bet $b^* = p$
$L \ge H(p)$ (Theorem 1.9)	$W \le \sum p \log o - H(p)$ (Theorem 1.11)
$\log m - H(p)$ bits saved per symbol	$W^* = \log m - H(p)$ at uniform odds

2 Inside information

A gambler who learns something about the race before betting, a tip, a weather report, the past form of the horses, can do better than the Kelly bet computed from the public probabilities.

A coder who learns something about the symbol before coding it can do better than the optimal code for the marginal distribution.

In both cases the gain is the mutual information between the side information and the outcome.

2.1 Worked examples

Example 2.1 *(A tip that is right nine times in ten).* Two horses, $p = (\frac{1}{2}, \frac{1}{2})$, even odds 2-for-1, so $W^*(X) = 1 - H(X) = 0$: without information there is nothing to win. A tipster names a horse Y and is right with probability 0.9: $p(x \mid y) = 0.9$ if $x = y$ and 0.1 otherwise. Then $H(X \mid Y) = h(0.1) = 0.4690$ bits and

$$I(X;Y) = H(X) - H(X \mid Y) = 1 - 0.4690 = 0.5310 \text{ bits.}$$

As a gambler. The conditional Kelly bet stakes 0.9 on the tipped horse and 0.1 on the other. When the tip is right, $S = 0.9 \cdot 2 = 1.8$ and $\log S = 0.8480 = i$; when it is wrong, $S = 0.2$ and $\log S = -2.3219 = i$. The doubling rate is $0.9(0.8480) + 0.1(-2.3219) = 0.5310 = I(X;Y)$. After 100 races the wealth is $2^{53.1} \approx 9.7 \times 10^{15}$ to first order in the exponent, from a starting stake of one unit and a bookmaker who quotes fair odds.
As a compressor. Encoder and decoder both see the tips Y^n. Given Y_k, the symbol X_k is "tip right" with probability 0.9 or "tip wrong" with probability 0.1, and the code describes 100 outcomes in fewer than $100h(0.1) + 2 = 48.9$ bits on average, against at least 100 bits for any code that does not see the tips: more than $51.1 = 100\, I(X;Y) - 2$ bits saved. Outcome by outcome, a right tip saves $1 - \log \frac{1}{0.9} = 0.848$ bits and a wrong tip costs $\log \frac{1}{0.1} - 1 = 2.322$ bits, the same two numbers as the wealth column.

References

[1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory.* Wiley-Interscience, Hoboken, NJ, 2nd edition, 2006. The course textbook. Chapters 5 and 6 are the sources.